

Ex. 2

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```
print(uuid::UUIDgenerate())
```

```
## [1] "cecd02de-52bd-43a4-a5e0-fceabdba3432"
```

Question 1

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)
commute_range <- max(commute) - min(commute)
class_width <- ceiling(commute_range / 5)
print(paste("Range:", commute_range))
```

```
## [1] "Range: 27"
```

```
print(paste("Class Width:", class_width))
```

```
## [1] "Class Width: 6"
```

```
breaks <- seq(min(commute), min(commute) + (5 * class_width), by = class_width)
freq_table <- table(cut(commute, breaks = breaks, right = FALSE, include.lowest = TRUE))
print("Frequency Distribution:")
```

```
## [1] "Frequency Distribution:"
```

```
print(freq_table)
```

```
##
## [18,24) [24,30) [30,36) [36,42) [42,48]
##      5      6      5      6      2
```

```
cum_freq <- cumsum(freq_table)
print("Cumulative Frequency:")
```

```
## [1] "Cumulative Frequency:"
```

```
print(cum_freq)
```

```
## [18,24) [24,30) [30,36) [36,42) [42,48]  
##      5      11      16      22      24
```

Question 2

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,  
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)  
library(ggplot2)  
df <- data.frame(commute)  
ggplot(df, aes(x = commute)) +  
  geom_histogram(bins = 5, fill = "steelblue", color = "white") +  
  labs(  
    title = "Distribution of Daily Commuting Times",  
    x = "Commuting Time (minutes)",  
    y = "Frequency"  
  )
```

